**Universidad de los Andes**
**Department of Computer and Systems Engineering**

# Graduation Project Proposal
**Bachelor's in Computing and Systems Engineering**

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# 1 Introduction and Motivation

When doing research or working in a field that involves other people's data it is crucial to set clear boundaries as to how that data is going to be collected, managed and distributed. Nowadays our most important asset is data, thus, it becomes a necessity to create tools that allow us to set such boundaries, one of those tools is differential privacy. Differential privacy describes a promise that is made by a data holder or curator to a data subject: "You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available".[1] In other words differential privacy is the process that either a data curator or a data holder must do in order to ensure that a single individual's data cannot be extracted from the study as a whole. This project focuses primarily on the concept of **Central differential privacy** which states that it is the responsibility of the curator (i.e a central party that receives all the raw data from the participants and adds privacy to them) to assure that the data is going to be differentially private.

On the other hand quantum information and computation is rapidly becoming mainstream as a response to the growing necessity of larger and faster computation. Quantum computing significantly advances computational capabilities, outperforming their classical counterparts in certain areas. Given the inherent noise of the current quantum framework, one could argue that it is a perfect scenario for differential privacy. In fact, the promise that the classical DP curator makes to the data holder will not change when in the quantum setting. However, the means that the curator will take to reach that promise do. The motivation for this project is to understand the literature that involves techniques of quantum differential privacy and contrast those techniques with experimental data collected in a quantum simulation experiment. With these results the project hopes to open the door to further studies in quantum differential privacy applications which as quantum computing keeps growing the results might just keep getting even better.

# 2 State of the art

It was important to figure out what the current investigative and academic situation was with regards to the fields of quantum information and computation and differential privacy. For this an investigation was made in which different databases were searched and analized to find out the amount of papers that covered the topics mentioned before. The investigation found 61 relevant papers that focused on the topics and the specific distributions of each one are compiled in Figure 1.

As shown in Figure 1 the investigation yielded positive results as there were many academic papers that not only mentioned the topic but gave valuable research and insight on possible angles the project could leap into. Among the most interesting ones are: [2] in which a hybrid quantum machine learning framework was proposed to allow differential privacy in the model. [3] which tested different quantum differential privacy mechanisms along with proposing its own **Lifted** quantum differential privacy, which enhanced the randomness induced in the quantum states. [4] in which an extended analysis was made to characterize the protection quantum noise can offer in the setting of quantum-classical algorithms. And finally in the study of Zhou and Ying [5] they define a framework of differential privacy in the quantum setting and examine how different noise mechanisms change the protection of a privacy budget $\epsilon$.

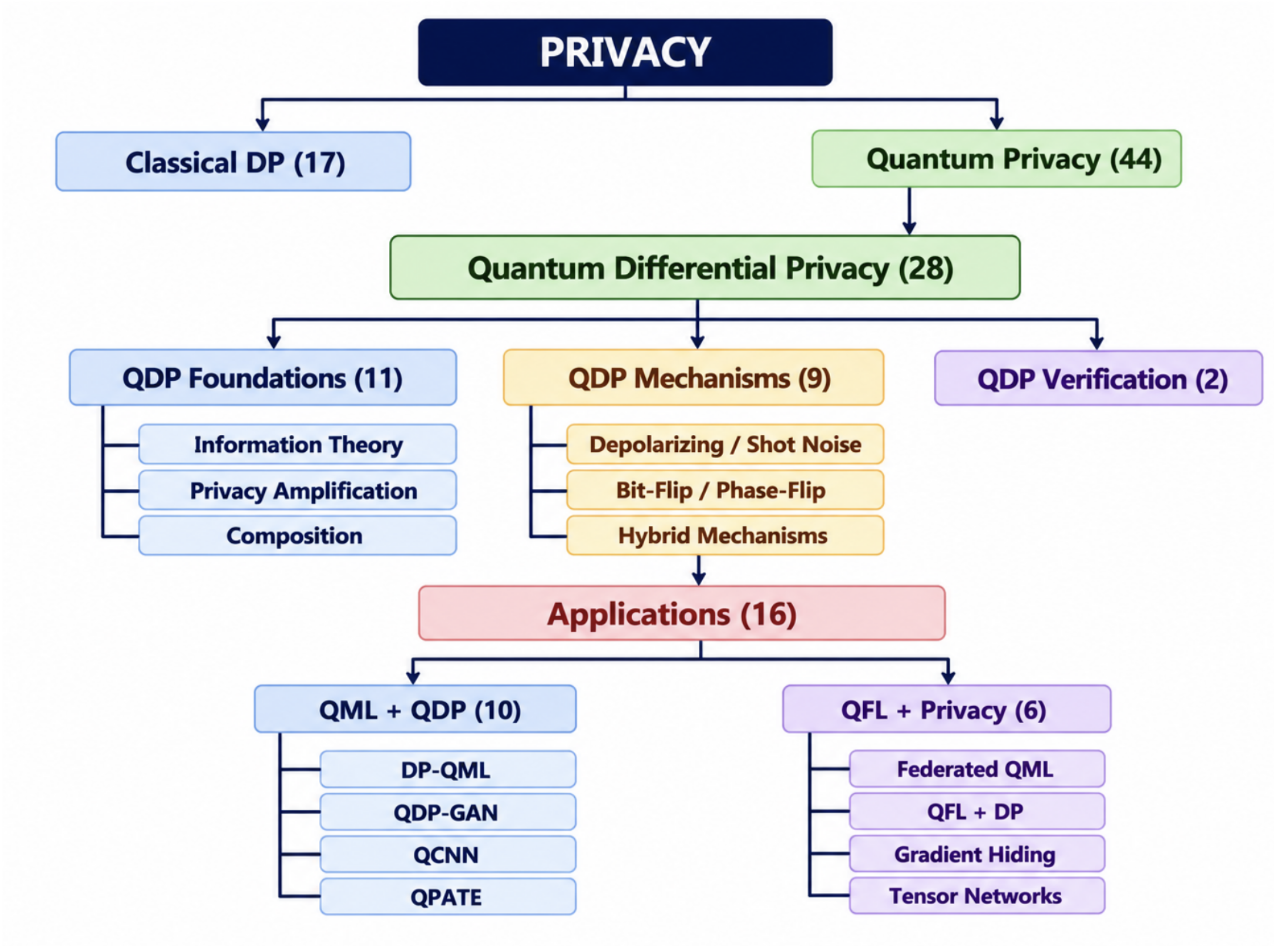


Figure 1: State of the art of quantum differential privacy. (own elaboration, based on the literature review described above)

# 3 Theorical Framework

## 3.1 Differential Privacy

**Definition 3.1.** *($\epsilon$, $\delta$)-Differential privacy [1]. A randomized algorithm $\mathcal{M}$ will provide ($\epsilon$, $\delta$)-Differential privacy (henceforth ($\epsilon$, $\delta$)-DP) if for any two outputs on two different databases D and D' differing by at most one record, and for all possible outputs A:*

$$Pr[\mathcal{M}(D) \in A] \leq e^{\epsilon} Pr[\mathcal{M}(D') \in A] + \delta$$

## 3.2 Quantum Information and Computation

**Definition 3.2.** *Quantum states [6]: A quantum state of the system X is described by a density matrix $\rho$ whose entries are complex numbers and whose indices (for both its rows and columns) have been place in correspondence with the classical state set $\Sigma$*

**Definition 3.3.** *Mixed state [5]: A mixed state of a quantum system is an ensemble*

$$\mathbb{E} = \{(p_1, |\psi_1\rangle), ..., (p_k, |\psi_k\rangle)\}$$

*which in turn can be represented by the density operator*

$$\rho_{\mathbb{E}} = \sum_i p_i |\psi_i\rangle \langle\psi_i|$$

*thus the resulting $\rho_{\mathbb{E}}$ will be the density matrix representation of the ensemble.*

**Definition 3.4.** *Quantum Channel [6]: A quantum channel describes a linear mapping over density matrices.*

Quantum channels are important as they are the generalization of an operation over a quantum state and in the case of this project we need to use operations such as the depolarizing channel which is defined thanks to this general definition.

**Definition 3.5.** *Trace distance [5]: It is the distance between two density operators and it is defined by:*

$$\tau(\rho, \sigma) = \frac{1}{2} Tr|\rho - \sigma|$$

*where $Tr(x)$ is the trace of the density matrix $x$*

**Definition 3.6.** *Quantum measurement [5]: It is the way to extract information from a quantum system. Mathematically, a measurement is modelled as a set of matrices $M = \{K_m\}$ with the condition of*

$$\sum_m K_m^\dagger K_m = I$$

*with $I$ being the identity of the density operator (i.e for a qubit $I = 2 \times 2$ identity matrix)*

When talking about quantum differential privacy below it won't really matter what the quantum state collapses into. Given this liberty we can define a POVM (Positive Operator-Valued Measure) which is a simplification over the measurement where we get a set of matrices $M = \{M_m\}$ with $M_m = K_m^\dagger K_m \; \forall m$ this is possible since we only care about the probabilities and so $p_m = Tr(K_m \rho K_m^\dagger) = Tr(M_m \rho)$

**Definition 3.7.** *Quantum noise [5]: It is an inherent physical process that occurs on the quantum state that causes it to change in some way.*

For the purposes of this investigation we are interested in three types of noise: General amplitude damping, Phase-amplitude damping and Depolarizing noise. Each one of these types represent a unique physical process which is summarized in greater detail in [3] and [5]. However for this proposal it is only necessary to note that these processes will randomize the quantum state by a certain degree.

### 3.3 Quantum Differential Privacy

**Definition 3.8.** *(Quantum differential privacy [5]). Let $d \in (0,1]$ and $\epsilon, \delta > 0$ be three constants. A quantum operation $\mathcal{E}$ is $(\epsilon, \delta)$-differentially private if for every POVM $M = M_m$, for all outputs $S \subseteq Out(M)$ and for all inputs $\rho, \sigma$ such that $\tau(\rho, \sigma) \leq d$, it holds that:*

$$Pr[\mathcal{E}(\rho) \in_M S] \leq e^{\epsilon} Pr[\mathcal{E}(\sigma) \in_M S] + \delta$$

There are many ways to obtain that epsilon in theory however Zhou and Ying proposed the following algorithm to obtain a quantity called proportional distance which can help obtain $\epsilon$.

**Definition 3.9.** *(Proportional Distance [5]). For two density operators $\rho$ and $\sigma$, their proportional distance is defined as:*

$$PD(\rho, \sigma) = \sup_{\{M_m\}, m} \max\left(\ln \frac{q_m}{p_m}, \ln \frac{p_m}{q_m}\right)$$

*where the supremum is over all POVMs $\{M_m\}$ and all possible outcomes $m$,*

$$p_m \equiv Tr[\rho M_m], \quad q_m \equiv Tr[\sigma M_m]$$

*are the probabilities of obtaining outcome $m$ when the measurement is performed on $\rho$, $\sigma$, respectively, and we make the conventions $\frac{0}{0} = 1$, $\frac{1}{0} = +\infty$, $\ln 0 = -\infty$, $\ln +\infty = +\infty$.*

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**Algorithm 1:** Computing $dPD$ [5]

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**Function** $dPD$ $(\rho, \sigma, \theta)$
`//` $\rho$ `and` $\sigma$ `are two density operators in Hilbert space` $\mathcal{H}$ `and` $\theta$ `is the desired accuracy.`
Let $n$ be the dimension of the Hilbert space $\mathcal{H}$.
Set real number $max = 0, \lambda = \infty$.
**while** $|max - \lambda| > \theta$ **do**
    $\lambda = max$
    $max = \max_i(\rho_{ii}/\sigma_{ii})$
    $\eta = \rho - max * \sigma$
    Diagonalize $\eta$: $\eta = PDP^{\dagger}$
    `//` $P$ `is unitary and` $D$ `is diagonal matrix.`
    $\rho = P^{\dagger} \rho P$
    $\sigma = P^{\dagger} \sigma P$
**end**
**return** $\ln max$

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## 4 Objectives

### 4.1 General Objective

The general objective of this project will be to compare and contrast the theoretical privacy budget derived in [5] with an experimental budget measured on a practical quantum simulation on the PennyLane framework provided by quantum noise mechanisms such as the depolarizing channel, phase-amplitude damping and GAD.

### 4.2 Specific Objectives

- Gain an understanding of the classical definition of differential privacy, as well as quantum information and computation to understand quantum differential privacy and be able to design and implement an experiment which is based on these topics.
- Create an experiment on the quantum simulation framework PennyLane which will test $(\epsilon, \delta)-DP$ over a real dataset [7]. These results will be explored in further detail in the project's paper.
- Compare and contrast the theoretical budgets proposed in [5] and [3] with the results of the experiments in the simulations for each different noise mechanism.

## 5 Description of the Experiment

The experiment to find $\epsilon_{experimental}$ will take the following steps shown in Figure 2:

Step 1. Encode the dataset $D$ and $D'$ (Which is $D$ with one participant (row) removed) with the following rules:

(a) For categorical variables a set angle will encode each of the possible categories.

(b) For non categorical variables each one will be encoded by dense angle encoding, thus giving two variables per qubit.

The result from this encoding process are going qubits which we will name $Q_D$ and $Q_{D'}$ respectively.

Step 2. Run $Q_D$ and $Q_{D'}$ from **Step 1** over a quantum circuit (i.e aggregation). We are going to call the density matrices that result from this process $\rho$ and $\sigma$ respectively.

Step 3. Insert quantum noise into $\rho$ and $\sigma$ resulting in $\rho'$ and $\sigma'$.

Step 4. Run Algorithm 1 on $\rho'$ and $\sigma'$ to get the $\epsilon_{experimental}$ for that specific run

Step 5. Run the **Steps 1 - 4** for all pairs of $D$ and $D'$ to get the maximum $\epsilon$ which will become $\epsilon_{experimental}$ for the whole dataset and compare against $\epsilon_{theoretical}$

## 6 Expected Results

1. Contrast the privacy budget that results from running the experimental setup with the theorical privacy budget found in the literature.
2. Develop the source code for the experimental setup in PennyLane with Python.
3. Write a scientific article in IEEE format. The information within this article should demonstrate my understanding of the topic as it will contain the whole process of investigation, creation and result analysis of this project.
4. Design a poster which will summarize the most important parts of the project.

# 7 Ethical Considerations

It has been concluded that the investigation will not have any ethical risk. The investigation will only deal with an adaptation [8] of the Statlog German Credit Data dataset [7], originally compiled by Prof. Hans Hofmann in 1994 which has a public CC BY 4.0 license which allows for the free use of the dataset as long as it is properly referenced.

# 8 Activity Calendar

| **Week #** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Activity** | | | | | | | | | | | | | | | | |
| Investigation on the viability of the project and scope | | | | | | | | | | | | | | | | |
| Write the proposal of the project | | | | | | | | | | | | | | | | |
| Search for the dataset to be used in the experiment and treat it | | | | | | | | | | | | | | | | |
| Write the introductory section of the paper (Introduction, Theoretical Framework) | | | | | | | | | | | | | | | | |
| Create the experiment in the PennyLane framework with at least one noise mechanism | | | | | | | | | | | | | | | | |
| Analysis of the first experiment's results and adjustment | | | | | | | | | | | | | | | | |
| Create the experiment with the remaining noise mechanisms | | | | | | | | | | | | | | | | |
| Write the first draft of the paper | | | | | | | | | | | | | | | | |
| Adjustments to the experiments with respect to the result of the analysis | | | | | | | | | | | | | | | | |
| Write the second draft of the paper | | | | | | | | | | | | | | | | |
| Write and design the other deliverables (Poster, Presentation, etc) | | | | | | | | | | | | | | | | |
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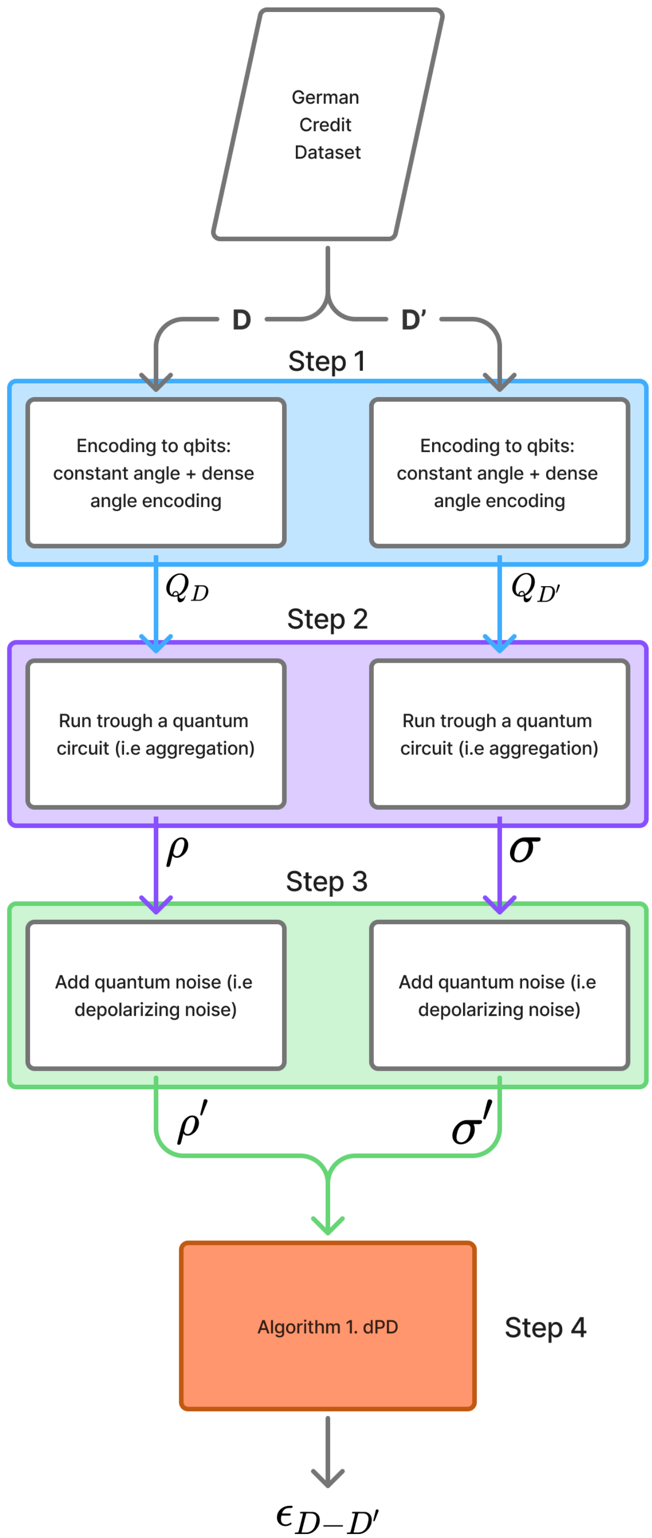


Figure 2: Diagram of the experimental setup for **Step 1 - 4**.